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EXPOSURE APPARATUS AND CONTROL METHOD THEREOF

Abstract

An exposure apparatus includes an exposure chuck and a wafer height fine-tuning device. The exposure chuck is used to fix a wafer, and the wafer height fine-tuning device is disposed between the exposure chuck and the wafer to adjust heights of a local area of the wafer. In addition, an exposure apparatus control method is also disclosed herein.

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Background/Summary

BACKGROUND

Field of Invention

[0001] The present disclosure relates to an exposure apparatus and a control method thereof. More particularly, the present disclosure relates to a semiconductor device exposure apparatus and a control method thereof.

Description of Related Art

[0002] Due to the rapid development of integrated circuits, minimizing the device dimension and increasing the integration level have become the mainstream in the semiconductor industry. Generally, a semiconductor device is fabricated by performing a series of processes including deposition processes, photolithography processes, etching processes, and ion implantation processes. The key technology to decide the critical dimension (CD) is in photolithography and etching. A typical photolithography process is conducted with a photolithography tool including a stepper or a scanner. The photolithography process normally includes coating a photoresist layer on a material layer to be patterned with a coater unit of the track, partially exposing the photo resist layer by the stepper or the scanner, post-exposure baking (PEB) the exposed photoresist layer with a PEB unit, and developing the exposed photoresist layer with a developer unit. Thereafter, an etching process is preformed to the material layer by using the developed photoresist layer as a mask, so as to transfer the patterns from the developed photoresist layer to the material layer.

[0003] However, as the exposure accuracy requirements are getting higher and higher, there is a need to improve the exposure accuracy so as to improve the quality and yield of semiconductor devices.

SUMMARY

[0004] The summary of the present invention is intended to provide a simplified description of the disclosure to enable readers to have a basic understanding of the disclosure. The summary of the present invention is not a complete overview of the disclosure, and it is not intended to point out the importance of the embodiments/key elements of the present invention or define the scope of the invention.

[0005] One objective of the embodiments of the present invention is to provide an exposure apparatus and a control method thereof to improve the quality and yield of the semiconductor devices.

[0006] To achieve these and other advantages and in accordance with the objective of the embodiments of the present invention, as the embodiment broadly describes herein, the embodiments of the present disclosure provides an exposure apparatus including an exposure chuck and a wafer height fine-tuning device. The exposure chuck fixes a wafer and the wafer height fine-tuning device is disposed between the exposure chuck and the wafer to adjust heights of a local area of the wafer.

[0007] In some embodiments, the exposure chuck includes an exposure chuck height adjustment device to adjust heights of the exposure chuck so as to further adjust heights of the wafer.

[0008] In some embodiments, the wafer height fine-tuning device includes a plurality of inverse piezoelectric devices directly contacting the wafer to respectively adjust the heights of the local area of the wafer.

[0009] In some embodiments, the size of the inverse piezoelectric devices is about 0.1 mm*0.1 mm to 5 mm*5 mm.

[0010] In some embodiments, an adjustment height of the inverse piezoelectric devices is about −100 nm to +100 nm.

[0011] In some embodiments, an adjustment height accuracy of the inverse piezoelectric devices is 1 nm.

[0012] In some embodiments, the exposure apparatus further includes a plurality of scanning sensors to detect heights of the local area of the wafer to obtain wafer height data.

[0013] In some embodiments, the exposure apparatus further includes a height analysis module connected to the scanning sensors to analyze the wafer height data to obtain an analysis result.

[0014] In some embodiments, the exposure apparatus further includes a chuck height control module connected to the height analysis module to adjust heights of the exposure chuck and the heights of the wafer according to the analysis result of the height analysis module.

[0015] In some embodiments, the exposure apparatus further includes a fine-tuning control module connected to the chuck height control module to respectively adjust heights of the inverse piezoelectric devices according to the analysis result of the height analysis module so as to respectively adjust the heights of the local area of the wafer.
[0016] According to another aspect of the present disclosure, an exposure apparatus control method including the steps of providing an exposure chuck and fixing a wafer on the exposure chuck. In addition, a wafer height fine-tuning device is further disposed between the exposure chuck and the wafer to adjust heights of a local area of the wafer.
[0017] In some embodiments, the exposure apparatus control method further includes steps of moving the wafer and detecting heights of the local area of the wafer.
[0018] In some embodiments, the exposure apparatus control method further includes utilizing an exposure chuck height adjustment device to adjust heights of the exposure chuck.
[0019] In some embodiments, the exposure apparatus control method further includes utilizing a wafer height fine-tuning device to adjust the heights of the local area of the wafer.
[0020] In some embodiments, the exposure apparatus control method further includes exposing the local area of the wafer.
[0021] In some embodiments, the wafer height fine-tuning device includes a plurality of inverse piezoelectric devices directly contacting the wafer to respectively adjust the heights of the local area of the wafer.
[0022] In some embodiments, the exposure apparatus control method further includes utilizing a plurality of scanning sensors to detect heights of the local area of the wafer to obtain wafer height data.
[0023] In some embodiments, the exposure apparatus control method further includes utilizing a height analysis module to analysis the wafer height data to obtain an analysis result.
[0024] In some embodiments, the exposure apparatus control method further includes utilizing a chuck height control module to adjust heights of the exposure chuck according to the analysis result of the height analysis module so as to further adjust the heights of the wafer.
[0025] In some embodiments, the exposure apparatus control method further includes utilizing a fine-tuning control module to respectively adjust heights of the inverse piezoelectric devices according to the analysis result of the height analysis module so as to respectively adjust the heights of the local area of the wafer.
[0026] Hence, the exposure apparatus and the control method thereof may effectively scan the heights of the local area of the wafer by the scanning sensors, adjust the heights of the exposure chuck and the heights of the entire wafer by the exposure chuck height adjustment device, and respectively adjust the local heights of the wafer by the wafer height fine-tuning device so as to effectively improve the precision and accuracy of semiconductor devices during the wafer exposure and further improve the quality and yield of the semiconductor devices.
[0027] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.
[0028] The features, aspects, and advantages of the present disclosure are better understood by referring to the following detailed description. It is noted that both the foregoing general description and the following detailed description are merely illustrative and are intended to provide further explanations of the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0029] The present disclosure is best understood from the following detailed description and the

accompanying figures. It is noted that the elements in the figures may not be drawn to meet the exact scale. Some elements may be drawn with increased or decreased sizes for clarity of the discussion.

[0030] FIG. **1** is a schematic diagram of an exposure apparatus according to an embodiment of the present disclosure; and

[0031] FIG. **2** is a schematic flowchart of an exposure apparatus control method according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0032] To make the description of the present disclosure more detailed and complete, explanatory descriptions of the aspects and specific implementations of the embodiments are provided below. It is not to limit the embodiments of the present disclosure to only one form. The embodiments of the present disclosure can combine or be substituted with each other under beneficial circumstances. Other embodiments may be appended without further description or explanation.

[0033] Furthermore, spatially relative terms, such as below and above, etc., may be used in the present disclosure to describe the relationship of one element or feature to another element or feature in the drawings. In addition to the orientation depicted in the figures, spatially relative terms are intended to encompass different orientations of the device in use or step. For example, the device may be otherwise oriented (e.g., rotated 90 degrees or otherwise) and the spatially relative terms of this disclosure are to be interpreted accordingly. In this disclosure, unless otherwise indicated, the same element numbers in different figures refer to the same or similar elements formed from the same or similar materials by the same or similar methods.

[0034] FIG. **1** is a schematic diagram of an exposure apparatus according to an embodiment of the present disclosure and FIG. **2** is a schematic flowchart of an exposure apparatus control method according to an embodiment of the present disclosure.

[0035] Referring to FIG. **1**, as shown in the drawing, the exposure apparatus **100** is utilized to form the required circuit structure on a wafer **130** during a wafer exposure process. The exposure apparatus **100** includes an exposure chuck **110** and a wafer height fine-tuning device **120**. The exposure chuck **110** is utilized to fix the wafer **130**. It is worth noting that the wafer height fine-tuning device **120** is disposed between the exposure chuck **110** and the wafer **130** to adjust heights of a local area of the wafer **130**.

[0036] In addition, the exposure chuck **110** includes an exposure chuck height adjustment device **114** to adjust heights of the exposure chuck **110** so as to further adjust heights of the wafer **130**.

[0037] In some embodiments, the exposure chuck **110** further includes a plurality of vacuum suction cups **112** disposed on the exposure chuck height adjustment device **114** to suck the wafer **130** so as to fix the wafer **130** on the exposure chuck **110** to further process the wafer **130**.

[0038] In some embodiments, the wafer height fine-tuning device **120** includes a plurality of inverse piezoelectric devices, e.g. a first inverse piezoelectric device **121**, a second inverse piezoelectric device **122**, a third inverse piezoelectric device **123**, etc. The inverse piezoelectric devices directly contact the wafer **130** to respectively adjust heights of corresponding parts of the local area of the wafer **130**.

[0039] In some embodiments, when appropriate voltages are applied, the first inverse piezoelectric device **121** and the second inverse piezoelectric device **122** push corresponding parts of the wafer **130** upward, and the third inverse piezoelectric device **123** moves downward, in conjunction with the downward force of the vacuum suction cups **112**, to cause a corresponding part of the wafer **130** to move downward, and therefore the heights of the corresponding parts of the local area of the wafer **130** are adjusted according to the measurement results of the surface height of the wafer **130** so that the exposure light may be accurately focused on the surface of the wafer **130**.

[0040] In some embodiments, the inverse piezoelectric devices of the wafer height fine-tuning device **120** are preferably distributed on the entire surface of the exposure chuck **110** to respectively and locally fine-tune the surface height of the wafer **130** fixed on the exposure chuck

110.

[0041] In some embodiments, the inverse piezoelectric devices of the wafer height fine-tuning device **120** and the vacuum suction cups **112** of the exposure chuck **110** are preferably staggered to effectively locally fine-tune the surface heights of the wafer **130**.

[0042] In some embodiments, the vacuum suction cups **112** of the exposure chuck **110** may be replaced with the electrostatic suction cups, without departing from the spirit and protection scope of the present disclosure.

[0043] In some embodiments, the size of the first inverse piezoelectric device **121**, the second inverse piezoelectric device **122** and the third inverse piezoelectric device **123** is about 0.1 mm*0.1 mm to 5 mm*5 mm, but not limited thereto.

[0044] In some embodiments, the adjustment height of the first inverse piezoelectric device **121**, the second inverse piezoelectric device **122** and the third inverse piezoelectric device **123** is about −100 nm to +100 nm, but not limited thereto.

[0045] In some embodiments, the adjustment height accuracy of the first inverse piezoelectric device **121**, the second inverse piezoelectric device **122** and the third inverse piezoelectric device **123** is 1 nm, but not limited thereto.

[0046] In some embodiments, the exposure apparatus **100** further includes a plurality of scanning sensors **140**, e.g. a first scanning sensor **141**, a second scanning sensor **142**, etc., to detect heights of the local area of the wafer **130** to obtain a plurality of wafer height data.

[0047] In some embodiments, the exposure apparatus **100** further includes a height analysis module **150** connected to the scanning sensors **140** to analyze the wafer height data so as to obtain an analysis result.

[0048] In some embodiments, the exposure apparatus **100** further includes a chuck height control module **160** connected to the height analysis module **150** to adjust the heights of the exposure chuck **110** so as to further adjust the heights of the entire wafer **130** according to the analysis result of the height analysis module **150**.

[0049] In some embodiments, the exposure apparatus **100** further includes a fine-tuning control module **170** connected to the chuck height control module **160** to respectively adjust the heights of the inverse piezoelectric devices so as to respectively adjust corresponding local parts of the local area of the wafer **130** according to the analysis result of the height analysis module **150**.

[0050] That is to say, the exposure apparatus **100** may adjust the entire heights of the wafer **130** by adjusting the heights of the exposure chuck **110** and adjust corresponding local parts of a local area of the wafer **130** by respectively adjusting the heights of the inverse piezoelectric devices to accurately adjust the surface heights of the wafer **130** so as to precisely focus thereon.

[0051] Simultaneously referring to FIG. **2** and FIG. **1**, an exposure apparatus control method **200** includes the following steps. In step **210**, an exposure chuck **110** is provided. In step **220**, a wafer **130** is fixed on the exposure chuck **110**. In addition, a wafer height fine-tuning device **120** is equipped between the exposure chuck **110** and the wafer **130** to adjust heights of the local area of the wafer **130**.

[0052] In some embodiments, the exposure apparatus control method **200** further includes a step **230** of moving the wafer **130**, and a step **240** of sensing heights of the local area of the wafer **130**. In addition, the wafer **130** is sucked on the exposure chuck **110** to horizontally move together with the exposure chuck **110**. Furthermore, the step **240** of sensing heights of the local area of the wafer **130** may utilize a plurality of scanning sensors **140** to respectively detect the heights of the local area of the wafer **130** to obtain a plurality of wafer height data. In some embodiments, a plurality of scanning sensors **140** includes a first scanning sensor **141**, a second scanning sensor **142**, etc., and the scanning sensors **140** may move relative to the wafer **130** to detect the surface heights of the wafer **130** before the wafer **130** is exposed.

[0053] In some embodiments, in step **250**, the exposure apparatus control method **200** may further include utilizing an exposure chuck height adjustment device **114** to adjust heights of the exposure

chuck **110** and simultaneously adjust the heights of the entire wafer **130**.
[0054] In some embodiments, in step **260**, the exposure apparatus control method **200** may further include utilizing a wafer height fine-tuning device **120** to respectively adjust heights of the local area of the wafer **130**.
[0055] In some embodiments, in step **270**, the exposure apparatus control method **200** may further include respectively exposing the local area of the wafer **130**. In addition, another local area of the wafer **130** is further detected, adjusted and exposed until the entire wafer **130** is exposed.
[0056] In some embodiments, a plurality of inverse piezoelectric devices respectively directly contact the wafer **130** to respectively adjust the heights of the local area of the wafer **130**.
[0057] In some embodiments, a plurality of scanning sensors **140** are equipped in front of the exposing element, e.g. exposing lens, to pre-detect heights of the local area of the wafer **130**. In addition, the exposure apparatus control method **200** utilizes a height analysis module **150** to analyze the wafer height data to obtain an analysis result. Furthermore, the analysis result is sent to the chuck height control module **160** and the fine-tuning control module **170**.
[0058] In addition, the chuck height control module **160** may adjust the heights of the exposure chuck **110** and further adjust the heights of the wafer **130** according to the analysis result of the height analysis module **150**. Furthermore, the fine-tuning control module **170** is utilized to further respectively adjust the heights of the inverse piezoelectric devices according to the analysis results of the height analysis module **150** to respectively adjust the local height of the corresponding local area of the wafer **130** so that the surface heights of the wafer **130** in the local area tend to be uniform to improve precision and accuracy during the exposure process.
[0059] Accordingly, the exposure apparatus and the control method thereof may effectively scan the heights of the local area of the wafer by the scanning sensors, adjust the heights of the exposure chuck and the heights of the entire wafer by the exposure chuck height adjustment device, and respectively adjust the local heights of the wafer by the wafer height fine-tuning device so as to effectively improve the precision and accuracy of semiconductor devices during the wafer exposure and further improve the quality and yield of the semiconductor devices.
[0060] The present disclosure is described in considerable detail with some embodiments. Other embodiments may be feasible. Therefore, the scope and spirit of the claims that are appended should not be limited to the description of the embodiments in the present disclosure.
[0061] For one skilled in the art, the present disclosure may be modified and changed as long as not departing from the spirit and scope of the present disclosure. If the modifications and changes are within the scope and spirit of the claims that are appended, they are covered by the present disclosure.

Claims

1. An exposure apparatus, comprising: an exposure chuck to fix a wafer; and a wafer height fine-tuning device disposed between the exposure chuck and the wafer to adjust heights of a local area of the wafer.
2. The exposure apparatus according to claim 1, wherein the exposure chuck comprises an exposure chuck height adjustment device to adjust heights of the exposure chuck so as to further adjust heights of the wafer.
3. The exposure apparatus according to claim 2, wherein the wafer height fine-tuning device comprises a plurality of inverse piezoelectric devices directly contacting the wafer to respectively adjust the heights of the local area of the wafer.
4. The exposure apparatus according to claim 3, wherein a size of one of the inverse piezoelectric devices is about 0.1 mm*0.1 mm to 5 mm*5 mm.
5. The exposure apparatus according to claim 4, wherein an adjustment height of one of the inverse piezoelectric devices is about −100 nm to +100 nm.

6. The exposure apparatus according to claim 5, wherein an adjustment height accuracy of the one of the inverse piezoelectric devices is 1 nm.

7. The exposure apparatus according to claim 3, further comprising: a plurality of scanning sensors to detect heights of the local area of the wafer to obtain a plurality of wafer height data.

8. The exposure apparatus according to claim 7, further comprising: a height analysis module connected to the scanning sensors to analyze the wafer height data to obtain an analysis result.

9. The exposure apparatus according to claim 8, further comprising: a chuck height control module connected to the height analysis module to adjust heights of the exposure chuck according to the analysis result of the height analysis module so as to further adjust the heights of the wafer.

10. The exposure apparatus according to claim 9, further comprising: a fine-tuning control module connected to the chuck height control module to respectively adjust heights of the inverse piezoelectric devices according to the analysis result of the height analysis module so as to respectively adjust the heights of the local area of the wafer.

11. An exposure apparatus control method, comprising: providing an exposure chuck; and fixing a wafer on the exposure chuck, wherein a wafer height fine-tuning device is further disposed between the exposure chuck and the wafer to adjust heights of a local area of the wafer.

12. The exposure apparatus control method according to claim 11, further comprising: moving the wafer; and detecting heights of the local area of the wafer.

13. The exposure apparatus control method according to claim 12, further comprising: utilizing an exposure chuck height adjustment device to adjust heights of the exposure chuck.

14. The exposure apparatus control method according to claim 13, further comprising: utilizing a wafer height fine-tuning device to adjust the heights of the local area of the wafer.

15. The exposure apparatus control method according to claim 14, further comprising: exposing the local area of the wafer.

16. The exposure apparatus control method according to claim 15, wherein the wafer height fine-tuning device comprises a plurality of inverse piezoelectric devices directly contacting the wafer to respectively adjust the heights of the local area of the wafer.

17. The exposure apparatus control method according to claim 16, further comprising: utilizing a plurality of scanning sensors to detect heights of the local area of the wafer to obtain a plurality of wafer height data.

18. The exposure apparatus control method according to claim 17, further comprising: utilizing a height analysis module to analysis the wafer height data to obtain an analysis result.

19. The exposure apparatus control method according to claim 18, further comprising: utilizing a chuck height control module to adjust heights of the exposure chuck according to the analysis result of the height analysis module so as to further adjust the heights of the wafer.

20. The exposure apparatus control method according to claim 19, further comprising: utilizing a fine-tuning control module to respectively adjust heights of the inverse piezoelectric devices according to the analysis result of the height analysis module so as to respectively adjust the heights of the local area of the wafer.
